

Refined SBERT: Representing sentence BERT in manifold space

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ABSTRACT

Distributed sentence representations have shown great power in a wide range of natural language processing (NLP) tasks. Meanwhile, a contextualized sentence representation, called Sentence BERT (SBERT), achieves excellent performance in quite a few NLP tasks. However, Sentence BERT is typically learned in the Euclidean space, the geometric structure of sentence representations and their relations to the representations of sentence's contexts has not been carefully studied yet. In this paper, we propose a new sentence representation method, named Refined SBERT, which utilizes manifold learning to refine sentence BERT by re-embedding sentence vectors from the original embedding space to a new refined semantic space. In order to map sentences to manifold space, we utilize neighborhood preserving embedding to construct the local manifold structure of the sentences. Our method can discover the local geometric structure and obtain a compact sentence BERT subspace, which can best detect the essential semantic structure. We conduct comprehensive experiments on various sentence embedding tasks including semantic textual similarity tasks, text classification and document clustering, and the experimental results show that the proposed model achieved promising results, comparing to several popular sentence representations.

1. Introduction

In many natural language processing (NLP) tasks, sentence representation is an important part of text representation. In the past decade, researchers have proposed different methods to generate sentence vectors [1–3]. A simple implementation of sentence representation is to sum or average the embedding of all its constituent words, such as bag of words model, Glove [4] and Word2vec [5]. These models have achieved good results in natural language processing tasks, but they ignore the order of words and lose the semantic information of the text to a certain extent.

In order to solve the above problems, researchers have proposed a series of distributed sentence representation methods. For example, Collobert et al. [6] use convolutional neural networks (CNN) to model sentences. This method uses the convolution operation of the CNN model to capture different levels of text features in parallel, and the pooling operation can capture the local features of the text. Yin et al. proposed a Bi-CNN-MI sentence embedding model [7]. This method is based on the DCNN model to extract four different granularities of

words, long N-Gram, short N-Gram, and sentences. Pagliardini et al. proposed a Sent2Vec [8], which uses n-gram embedding and word vectors to form the vector representation of sentences. The above methods have good applications in sentence representation tasks. However, they are all based on static word representation models and then trained by neural networks to obtain sentence vector representations. Recently, in view of the fact that the static word representation model cannot solve the polysemy problem of a word in different contexts, researchers have proposed a dynamic word representation model to select different word meaning information according to the context information. For example, in the recently proposed ELMO [9] model and BERT [10] model, each word in a sentence can obtain three different embedding representations in the pre-training network of the ELMO model, and the word embedding accumulated using different weights can be used as the final feature of the word. Devlin et al. [10] proposed the BERT model. BERT not only uses a two-way language model, but also uses Transformer as a feature extractor. BERT trains the model through a large enough corpus, which can capture rich semantic information, and trains the model through a large number of unlabeled data to obtain text

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Table 1
Sentence Similarity.

Sentences	Similarity	Source
(Sentence1, Sentence2)	0.6	Real score
(Sentence3, Sentence4)	1.0	
(Sentence1, Sentence2)	0.74	SBERT
(Sentence3, Sentence4)	0.39	
(Sentence1, Sentence2)	0.15	Our method
(Sentence3, Sentence4)	0.28	

representation. We can fine tune BERT for different tasks to obtain text representations that meet the needs of different tasks. Based on the above advantages, BERT has good applications in many natural language processing tasks. Recently, Reimers et al. [11] studied the regression task of BERT in sentences such as clustering and semantic search. Reimers et al. found that although BERT achieved good results on sentence-pair regression tasks such as text semantic similarity. However, BERT requires both sentences to be input into the network, resulting in huge overhead. Moreover, BERT is also not suitable for semantic similarity search and unsupervised tasks, such as clustering tasks. In order to solve this problem, Reimers et al. used Siamese network structure to input different sentences into two BERT models, and proposed Sentence BERT (SBERT). Then, the word vectors output by the BERT model are averagedly pooled to obtain fixed-length, local sentence vectors with rich semantic information. Sentence BERT (SBERT) can reduce the training time of BERT from 65 h to 5 s while ensuring accuracy.

Despite Although distributed sentence representation methods have shown promising results in natural language processing tasks, the geometric structure of sentences has not been thoroughly explored. These methods typically map sentences to low-dimensional dense vectors that exist in Euclidean space as points. Sentences with similar semantics are represented by vectors that are closer in Euclidean space, while those with opposite semantics are represented by vectors that are farther apart. However, current sentence representation methods do not fully consider the geometric structure of sentences. Consequently, the similarity of sentences in Euclidean metric space may not align with human judgment. For instance, let us consider four sentences from the STS13¹ dataset: “Saudi gas truck blast kills at least 22,” “Nigeria church blast kills at least 12,” “French election turnout higher than expected,” and “French presidential election: first round - live updates.” The true score of sentence 1 and sentence 2 is ‘0.6,’ while the score of sentence 3 and sentence 4 is ‘1.’ We obtain the vector representation of these sentences using the SBERT pre-training model and measure their similarity using cosine distance, as shown in Table 1.

Table 1 displays the actual scores between (Sentence1, Sentence2) and (Sentence3, Sentence4) sentences, as rated by humans. Additionally, Table 1 presents the similarity of sentence vectors obtained through SBERT pre-training model in Euclidean metric space. From Table 1, it is apparent that the sentence similarity of the sentence vector obtained by SBERT is completely opposite to the real scores of sentence pairs. To address this inconsistency between sentence semantic similarity and Euclidean space, this paper introduces manifold learning to distributed word representation. Manifold learning involves describing the local geometric structure information between sentence vectors by constructing an adjacency graph structure of sentence vectors in high-dimensional space. This process “tiles” the sample distribution group in the high-dimensional feature space to a low-dimensional space [12–16]. The sample distribution in the original space may be distorted, but after tiling, it becomes more conducive to measuring the distance between word vectors. As a result, the distance between sentence vectors better reflects their similarity.

Motivated by the discussion above, we proposed an unsupervised method called Refined SBERT, to learn manifold sentence representation. Our method aims to discover the local geometrical structure of the sentence space. To map sentences to a manifold space, we employ neighborhood preserving embedding (NPE) [17] to construct adjacency graphs of sentences on the local structure. This approach helps to describe the geometric relationship between sentences, discover the potential manifold structure between sentences, and make the semantic information and relations between sentences consistent, thereby improving the results of downstream tasks. We applied manifold learning to the SBERT model to obtain the similarity between the (Sentence1, Sentence2) and (Sentence3, Sentence4) sentences, as shown in Table 1. The results show that the sentence vector similarity obtained by our method is consistent with the true score of the sentence pairs. It can be concluded that the second sentence pair (Sentence3 and Sentence4), which mentions the same event, the French election, is expected to be more similar than the first (Sentence1 and Sentence2), where both sentences refer to different events taking place at different times and locations. Thus, the similarity scores demonstrate that the proposed method improves upon SBERT to learn the correct ordering of semantic similarities between sentences.

In summary, our contributions are presented as follows:

- (1) We propose a new sentence representation method, named Refined SBERT, which analyzes deep contextualized models from the perspective of sentence manifold structure.
- (2) Sentence BERT is typically learned in the Euclidean space, and our approach considers the problem of sentence embeddings on the sentence manifold. It models the sentence space as a sub-manifold embedded in the ambient space.

2. Related work

2.1. Sentence representations

In recent years, several distributed sentence representations methods have been proposed. Kiros et al. proposed an unsupervised learning method for general distributed sentence coder [18]. They used continuous sentences in books to train the coder decoder model. When the model training is completed, the coder can encode new sentences into sentence vectors. Mikolov proposed the Doc2Vec method [19]. Doc2Vec can comprehensively examine the contextual semantics and contextual information of words, sentences and even paragraphs. Wieting et al. used the extra knowledge of the paraphrase database to learn general sentence embeddings [15]. Le et al. proposed the paragraph vector [16], which is an unsupervised method that can learn fixed-dimensional feature representation from variable-length sentences or paragraphs. Wang et al. introduced several attention mechanism methods to improve sentence embedding performance [17]. Tai et al. [20] integrated syntactic dependency and phrase structure features into the LSTM model, and used the tree LSTM to better express sentence semantics. Zhu et al. [21] proposed a recurrent convolutional neural network model, which introduced a convolutional layer and a sampling layer on the recurrent neural network model to better extract combined features.

Recently, in order to solve the problem of polysemy and choose different semantics considering the context, dynamic word embedding methods provide research ideas and directions. In 2018, Peters et al. first proposed the ELMo model. On this basis, the researchers proposed models such as BERT, RoBERTa, SBERT, and so on. BERT adopts Transformer as feature extractor. RoBERTa [22] uses dynamic masks: new mask patterns are generated each time a sequence is input to the model. SBERT uses the Siamese network structure to input different sentences into two BERT models, and introduces a pooling operation on the output of the BERT model, so that a fixed-length sentence vector can be generated. Wang et al. [23] study the layer-wise pattern of the word representation of deep contextualized models. Then, they proposed a

¹ <https://github.com/facebookresearch/SentEval>.

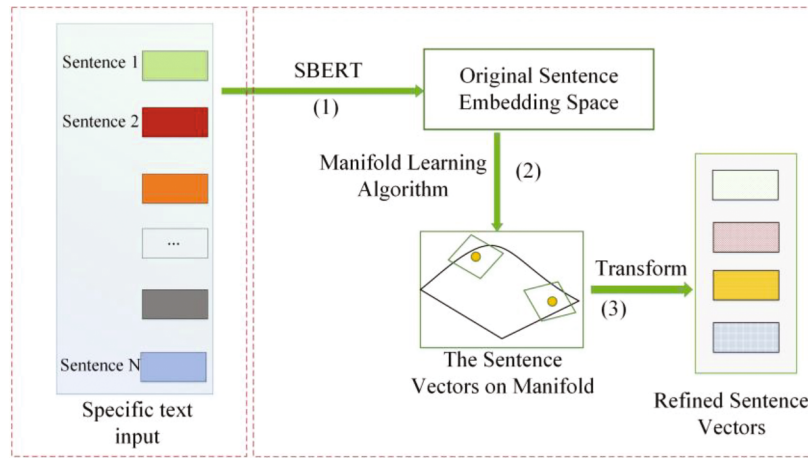


Fig. 1. The framework of our method.

new sentence embedding method by dissecting BERT-based word models through geometric analysis of the space spanned by the word representation. Since Transformer-based models are unable to process long sequences due to their self-attention operation, which scales quadratically with the sequence length. To address this limitation, Beltagy et al. [24] introduce the Longformer with an attention mechanism that scales linearly with sequence length, making it easy to process documents of thousands of tokens or longer. Gao et al. [25] proposed a sentence representation method based on contrastive learning to improve the embedded representation of sentence vectors. SimCSE first describe an unsupervised approach, which takes an input sentence and predicts itself in a contrastive objective, with only standard dropout used as noise. The difference between the method proposed method and the above method is that, Our method aims to leverage manifold learning to explore the inherent manifold structures of high-dimensional sentence vectors.

2.2. Manifold space models

Previous studies have shown that manifold learning is a better choice in preserving the local geometric structure of data and describing the similarity between the data [13]. For example, when the document tf-idf vector is used as a feature of document clustering and classification, manifold learning can discover the potential low-dimensional manifold structure between different documents, and explore the inherent relationship between different documents, which is helpful improve document performance in clustering and classification tasks. Manifold learning builds a nearest neighbor graph to model the local sentence manifold structure. It is natural to assume that two sufficiently close sentences have similar probability distribution. Through the above, the similarity between different sentences can be explored, so as to be consistent with the real semantics of the sentence. Recently, manifold learning also shows great effectiveness in deep learning. Tang et al. [26] explored the flow pattern information of the sample space from the complexity and similarity, and used an adaptive penalty to match the sample complexity with the network complexity. Tang et al. first introduced a mechanism for judging the complexity of the input image and appropriately adjusting the penalty for sparse filters. Secondly, the similarity between samples is preserved in the pruning results. Qi et al. [27] modeled the local manifold around the data points without encoder, and then used the local generative adversarial network to approximate the Laplace norm. For graph convolutional networks (GCN) [28] only fusion of static structural information, it is difficult to ensure that its structural information is optimal in the training process. Liu et al. [29] proposed a manifold dynamic graph convolution network (MRDGCN). Motivated by the success of these models, we introduce

manifold learning into sentence representation.

3. Our method

3.1. Overview of the proposed method

In this section, we describe the details of our proposed method. Our proposed sentence representation refinement method is based on a universal framework. The idea is to utilize manifold learning algorithms to re-embed sentence vectors from the original embedding space to a refined new embedding space. The framework is shown in Fig. 1. In this paper, we employ the sentence BERT as our sentence representations. Firstly, we get the sentence vectors corresponding to the document from the trained sentence BERT. Secondly, we use NPE to model the sentence vectors, use local adjacency graph to construct the neighborhood structure of the sentence vectors, and describe the internal rules between different sentences through the neighborhood structure of the sentence vectors. Finally, we optimize the NPE method and transform the sentence vectors of the original space into the manifold space to get a new sentence representation.

3.2. Sentence representation refinement by manifold learning

Due to the current distributed sentence representation methods ignore sentence geometry in the training process, it cannot accurately measure the similarity between sentence representations and affect the results of downstream tasks. To solve this problem, inspired by manifold learning, we introduce manifold learning algorithm into distributed sentence representation. Manifold learning aims to discover the inherent regularity of the distribution of high-dimensional data sets. The basic idea is: the points in the high-dimensional observation space are expanded into a manifold by the joint action of a few independent variables in the observation space. If we can effectively expand the curly manifold of the observation space or find the internal main variables, we can re-embed the sentence. Next, we use manifold learning algorithm neighborhood preserving embedding (NPE) [13] to re-embed the sentence BERT. NPE is a geometric property based on local linearity in data space, that is, assuming that the data sample point and its neighborhood point are in a linear or linear approximate manifold structure, then the data point can be represented by a linear combination of its leading points. NPE can find a suitable linear projection matrix to keep the local neighborhood structure between the corresponding projection samples in the low dimensional space. The following introduces the distributed sentence re-embedding model based on NPE.

For a given document, we first use sentence BERT to vectorize the document to obtain the vector representation of the sentence. Suppose

the set of sentence vectors is $S = \{s_1, s_2, \dots, s_N\}$, $s_i \in R^D$, N is the number of data, D represents the dimension of the data set, the projection transformation matrix $Sentence(X) \rightarrow Sentence(Y)$ reduces the sentence vectors from the original space R^D to the low-dimensional subspace R^d , $Y = \{y_1, y_2, \dots, y_N\} \in R^{d \times N}$ is the low-dimensional coordinate representation of the sentence vector sets S . The solution process of the distributed sentence re-embedding model based on NPE is as follows:

(1) Constructing an adjacency graph: suppose that G represents a graph of N nodes, and the i -th node corresponds to the data sample point s_i . We use k nearest neighbors to construct the adjacency graph: if s_j represents the nearest neighbors of s_i , then j has a direct edge from node to node i ;

(2) Calculating the weight matrix: the neighborhood relationship determined by step 1, $s_i = \sum_{s_j \in N(s_i)} w_{ij} s_j$ represents the neighborhood of the sample, w_{ij} is the weight value between s_i and s_j , W is the weight matrix, NPE is based on the assumption that each sample and its neighborhood points in the data space are in a linear or nearly linear manifold structure, so each sample point can be expressed linearly by a neighborhood point. The weight matrix is obtained by minimizing the reconstruction error

$$\min \sum_{i=1}^N \left\| s_i - \sum_{s_j \in N(s_i)} w_{ij} s_j \right\|_2^2 \quad (1)$$

$$s.t. \sum_{j=1}^k s_j = 1$$

where: s_j is the k nearest neighbors of s_i ; w_{ij} is the weight between s_j and s_i .

(3) Computing low dimensional embedding: NPE tries to ensure that if the original spatial sample point s_i can be represented linearly by the weight of the neighborhood points, then the sample point y_i projected to the low dimension can also be represented linearly by the same weight value. We use $y_i = U^T s_i$ to project s_i , and the low dimension of s_i is expressed as y_i and U is the projection transformation matrix. The optimal projection can be learned by minimizing the following cost function:

$$\left\{ \begin{array}{l} \min \sum_{i=1}^N \left\| y_i - \sum_{y_j \in N(y_i)} w_{ij} y_j \right\|_2^2 \\ s.t. YY^T = I \end{array} \right. \quad (2)$$

Where I is the identity matrix, $Y = \{y_1, y_2, \dots, y_N\}$.

The minimization formula (2) is to ensure that the weight values of the local neighborhood structure of the original spatial data points are shared in the low dimensional space. Take $y_i = U^T s_i$ into (3)

$$\begin{aligned} & \min \sum_{i=1}^N \left\| y_i - \sum_{y_j \in N(y_i)} w_{ij} y_j \right\|_2^2 \\ &= \sum_{i=1}^N \left\| U^T \left(s_i - \sum_{s_j \in N(s_i)} w_{ij} s_j \right) \right\|_2^2 \\ &= Tr(U^T SMS^T U) \end{aligned} \quad (3)$$

where $M = (I - W)^T (I - W)$, Tr is the trace of a matrix, The model of NPE can be expressed as follows:

$$\left\{ \begin{array}{l} \min = Tr(U^T SMS^T U) \\ s.t. U^T SS^T U = I \end{array} \right. \quad (4)$$

Formula (4) is solved by Lagrange multiplier method, and the problem of solving projection vector is transformed into the problem of finding its generalized eigenvalue

$$SMS^T U = \lambda SS^T U \quad (5)$$

The projection transformation matrix U is composed of the eigenvectors corresponding to the first d minimum non-zero eigenvalues of formula (5), and then the re-embedding representation of Sentence BERT can be obtained through $y_i = U^T s_i$.

Firstly, we use sentence BERT to vectorize the document to get the vector representation of sentences. Secondly, according to the formula (1) - (5), we model the manifold of the set of sentence vectors S . Finally, we use the adjacency weight matrix between the sentence vectors to describe the local geometric structure, so as to restore the real semantic measurement relationship between sentences. Algorithm 1 gives the process of our method.

Algorithm 1: Sentence BERT with manifold learning

Input: original sentence embedding S , sentence vectors $S = \{s_1, s_2, \dots, s_N\}$
Output: Refined sentence vectors $Y = \{y_1, y_2, \dots, y_N\} \in R^{d \times N}$ in manifold space
1: **For** all $S = \{s_1, s_2, \dots, s_N\}$ **do**
2: Construct an adjacency graph of sentence vectors through step (1)
3: Solve the adjacency weight matrix of the sentence vectors by formula (2)
4: According to formulas (2), (3), (4), (5), calculate the re-embedded representation of the sentence in the manifold space $sentence(s_i) \rightarrow sentence(y_i)$
5: **end for**
6: Return refined sentence vectors $Y = \{y_1, y_2, \dots, y_N\} \in R^{d \times N}$

4. Experiments

In this section, we present several experiments to verify the effectiveness of our proposed approach. We use the sentence vectors trained by the Sentence BERT model as the original input, It should be noted that, in the experiment, Sentence BERT without using any specific training data to fine tune the model. Therefore, We use an unsupervised Sentence BERT. This embedding choice is motivated by its state-of-the-art performance, which provides a strong base to learn the mappings.

4.1. Evaluation tasks and metrics

We test the baseline and proposed models on 15 standard evaluation benchmarks, covering three different tasks:

- (i) Semantic Textual Similarity Tasks: We use 2012–2016 STS datasets.² They contain sentence pairs and labels between 0 and 5, which indicate their semantic relatedness. Semantic relatedness is a special kind of similarity task, and we use the SICK-R [30] and the STS Benchmark dataset [31] in our experiments.
- (ii) Text Classification Tasks: Aiming to measure sentence embeddings' transfer capability to downstream tasks. We use 7 classification datasets³ for experiments, the dataset contains: MR: Binary sentiment prediction on movie reviews [32]. CR: Binary sentiment prediction on customer product reviews [33]. SUBJ: Binary subjectivity prediction on movie reviews and plot summaries [34]. MPQA: Phrase-level opinion polarity classification [35]. SST2: Stanford Sentiment Treebank with binary labels [36]. TREC: Question type classification with 6 classes [37]. MRPC: Microsoft Research Paraphrase Corpus for paraphrase prediction [38]. The classification accuracy is used as an evaluation metric.
- (iii) Document Clustering task: We use 20 Newsgroups datasets⁴ to conduct experiment. The dataset contains around 18,000 newsgroup documents partitioned into 20 classes.

4.2. Baseline models

We compare our method with state-of-the-art sentence embedding methods. These method include average of GloVe word embeddings, average of fast-text word embedding [39], InferSent-GloVe [40], Avg.

² <https://github.com/facebookresearch/SentEval>.

³ <https://github.com/saurabh3949/Text-Classification-Datasets>.

⁴ <https://qwone.com/~jason/20NewsGroups/>.

Table 2

Spearman rank correlation on textual similarity evaluation, bold represents the best result for each column.

Models	STS12	STS13	STS14	STS15	STS16	STSb	SICK-R
Avg. GloVe embeddings	55.14	70.66	59.73	68.25	63.66	58.02	53.76
Avg. BERT embeddings	38.78	57.98	57.98	63.15	61.06	46.35	58.40
BERT CLS-vector	20.16	30.01	20.09	36.88	38.08	16.50	42.63
InferSent-GloVe	52.86	66.75	62.15	72.77	66.87	68.03	65.65
BERT _{base} -flow	58.40	67.10	60.85	75.16	71.22	68.66	64.47
BERT _{base} -whitening	57.83	66.90	60.90	75.08	71.31	68.24	63.73
IS-BERT _{base}	56.77	69.24	61.21	75.23	70.16	69.21	64.25
CT-BERT _{base}	61.63	76.80	68.47	77.50	76.48	74.31	69.19
RoBERTa _{base} (first-last avg.)	40.88	58.74	49.07	65.63	61.48	58.55	61.63
RoBERTa _{base} -whitening	46.99	63.24	57.23	71.36	68.99	61.36	62.91
DeCLUTR-RoBERTa _{base}	52.41	75.19	65.52	77.12	78.63	72.41	68.26
SimCSE-BERT _{base}	68.40	82.41	74.38	80.91	78.56	76.85	72.23
SimCSE-RoBERTa _{base}	70.16	81.77	73.24	81.36	80.65	80.22	68.56
SBERT-base	70.97	76.53	73.19	79.09	74.30	77.03	72.91
SBERT-large	72.27	78.46	74.90	80.99	76.25	79.23	73.75
Refined SBERT-base	73.19	79.71	73.40	79.13	75.08	78.62	74.58
Refined SBERT-large	72.21	80.46	77.60	82.41	76.20	79.47	74.12

Table 3

Performance on text classification tasks, bold represents the best result for each column.4.3.

Models	MR	CR	SUBJ	MPQA	SST2	TREC	MRPC
Avg. GloVe embeddings	77.25	78.30	91.17	87.85	80.18	83.00	72.87
Avg. fast-text embeddings	77.96	79.23	91.68	87.81	82.15	83.60	74.49
Doc2Vec	78.02	83.51	92.85	88.49	82.31	86.84	75.29
Longformer	82.61	88.26	93.18	87.13	86.17	89.47	76.92
Avg. BERT embeddings	78.66	86.25	94.37	88.66	84.40	92.80	69.45
BERT CLS-vector	78.68	84.85	94.21	88.23	84.13	91.40	71.13
InferSent-GloVe	81.57	86.54	92.50	90.38	84.18	88.20	75.77
SBERT-base	83.64	89.43	94.39	89.86	88.96	89.60	76.00
SBERT-large	84.88	90.07	94.52	90.33	90.66	87.40	75.94
Refined SBERT-base	83.79	90.61	95.11	90.64	89.49	89.71	77.43
Refined SBERT-large	85.97	92.31	95.62	91.12	91.22	92.27	77.35

Table 4

Document clustering evaluation on the 20 Newsgroup datasets, bold represents the best result for each column.

Models	MI	NMI	ARI	Purity
Avg. GloVe embeddings	1.0444	0.3558	0.1797	0.3327
Avg. fast-text embeddings	0.5491	0.1886	0.0708	0.2057
Doc2Vec	0.6715	0.2274	0.0814	0.2149
Longformer	0.8825	0.2692	0.1825	0.3174
Avg. BERT embeddings	0.7320	0.2520	0.1030	0.2380
BERT CLS-vector	0.1056	0.0361	0.0104	0.1020
SBERT-base	0.9659	0.3255	0.1745	0.3540
SBERT-large	0.9412	0.3166	0.1590	0.3330
Refined SBERT-base	1.0991	0.3728	0.1911	0.3851
Refined SBERT-large	1.1048	0.3759	0.2043	0.4132

BERT embeddings: Average the last layer BERT word embedding, BERT-CLS-vector: Using BERT-[CLS] token embedding as sentence embeddings, Doc2Vec[19], Longformer[24], RoBERTa [22], SimCSE-BERT[], and post-processing methods such as BERT-flow [41] and BERT-whitening [42]. IS-BERT [43], which maximizes the agreement between global and local features, DeCLUTR [44] which takes different spans from the same document as positive pairs, CT-BERT [45], which aligns embeddings of the same sentence from two different encoders.

4.3. Experiments settings

Firstly, we input the sentences contained in semantic textual similarity, text classification and document clustering into sentence BERT to get the vector form of sentences. Then use the local preserving embedding to re-embed the sentence vectors. The value range of the neighborhood of locally preserving embedding is [50, 400]. For semantic textual similarity, we use the cosine similarity between sentence pairs as

the similarity score and report the spearman rank correlation coefficient. Table 2 shows the experimental results of different sentence embeddings. For the classification experiment, we use logistic regression as the classification algorithm. This is because logistic regression is a nonparametric method. The classification results of logistic regression directly reflect the capturing degree of sentence semantics by different sentence embedding methods (that is, whether the sentences from the same semantic class are embedded closer). The classification accuracy is used as the evaluation metric, and the experimental results of different methods are given in Table 3. For clustering experiments, we use the 20 Newsgroups datasets for clustering experiments, K-Means as the clustering method, and use Mutual Information (MI), Normalized Mutual Information (NMI), Adjusted Rand Index (ARI), and Purity as evaluation metrics. Table 4 shows the experimental results of different sentence embedding.

4.4. Results analysis

Table 2 shows the experimental results of different sentence embedding methods on semantic similarity task. According to the analysis of the data in Table 1, the experimental results of our method Refined SBERT-base and Refined SBERT-large on multiple datasets are better than those of other algorithms. It can be seen from Table 1 that the experimental results of Avg.BERT embedding and BERT CLS-vector are lower than those of Avg.Glove embedding, which also shows that in some cases, the experimental results of sentence vectors obtained by weighted average based on static word embedding method will achieve good results. Both Avg.BERT embeddings and Avg.Glove embeddings get sentence vectors by weighted average of the word vectors contained in the sentence, BERT CLS-vector using BERT-[CLS] token embedding as sentence embeddings. The above three methods do not use neural networks to train sentences, so the experimental results are lower than the

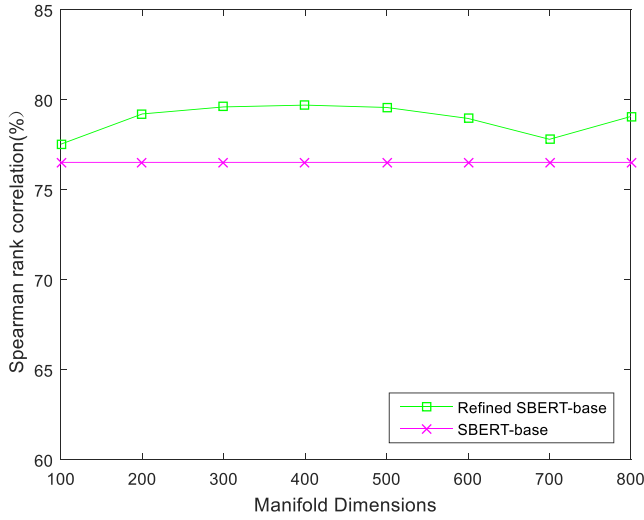


Fig. 2. Experimental results under different dimensions.

methods such as SBERT-base and SBERT-large. In Table 2, we compare the improved methods based on BERT, such as BERT_{base-flow}, CT-BERT_{base}, and SimCSE-BERT_{base}. From the experimental results given in Table 1, we can observe that our method significantly outperforms BERT_{base-flow}, CT-BERT_{base} on multiple datasets.

Table 3 shows the experimental results of this method and other methods on seven datasets. It is noted that, we use logistic regression as the classification algorithm. Since logistic regression is a non-parametric method, the performances of logistic regression directly reflect how well the topology of the embedding space captures sentence-level semantics (i.e., whether documents from the same semantic class are embedded closer). It can be seen from Table 3 that our method Refined SBERT-base and Refined SBERT-large have achieved good experimental results, and the experimental results are the best compared with other 7 sentence embedding methods. The experimental results of Avg. BERT embeddings and BERT CLS-vector on multiple datasets are better than the experimental results of Avg. GloVe embeddings and Avg. fast-text embeddings, which also verify that the BERT model generally performs better on supervised tasks. InferSent-GloVe is a model that uses neural network to train sentences. InferSent-GloVe uses SNLI corpus to train the representation of general sentence vectors. From Table 3 we can also see that the experimental results of InferSent-GloVe are better than Avg. GloVe embeddings and Avg. fast-text embeddings. The experimental results of SBERT on multiple datasets are better than methods such as Avg. GloVe embeddings, Avg. fast-text embeddings and InferSent-GloVe. This is because SBERT uses siamese and triplet objective function to obtain semantically meaningful sentence embedding.

Table 4 shows the clustering results of different methods on the 20 Newsgroup datasets. It is shown that feature quality is generally more important than clustering algorithms for document clustering tasks. Compared with other methods, our method has achieved the best experimental results. This demonstrates that directional similarity on sentence-level features is also beneficial for clustering tasks, which can be captured intrinsically in the manifold space. Experimental results on semantic similarity, text classification and document clustering show the effectiveness of our method. Our method is compared with the weighted average method and the neural network method respectively. Our method has achieved significant improvement on the three tasks. Our method constructs the local geometric structure between the sentence vectors through the nearest neighbor graph theory, and describes the similarity measurement between sentences. Through the above process, the re-embedding representation of sentences obtained by our method is consistent with the real semantic information between sentences.

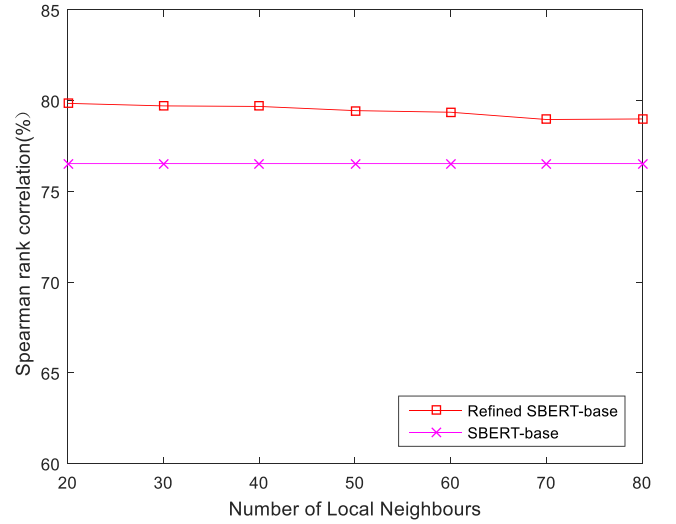


Fig. 3. Experimental results under different number of local neighbors.

4.5. Method discussion

In this section, we study two important parameters (the number of dimensions and domains included in NPE algorithm) that effect our method. We reduce the sentence vector to different dimensions to observe the experimental results of our method. At the same time, we use different sizes of neighborhood to observe the experimental results of our method.

4.5.1. Effect of dimensionality

To investigate the effects of dimensionality, we use NPE algorithm to reduce the sentence vector to different dimensions, and experiment on semantic textual similarity tasks.

The experimental results on STS13 are shown in Fig. 2. We can observe that our method is better than SBERT base in the range of dimensions from 100 to 800, which indicates that our method has good stability when dimensions change.

4.5.2. Effect of the number of local neighbours

In this section, we explore the influence of the number of different neighborhoods on the NPE algorithm, and conduct experiments on the semantic textual similarity data set STS13, and compare it with SBERT-base.

Fig. 3 shows the experimental results of the proposed method on the semantic textual similarity dataset STS13 with different number of local neighbors. We can also observe that with the increase of local neighbors, the experimental results of our method fluctuate less and are better than the experimental results of SBERT-base. Through the experimental results in Fig. 3, we can select the optimal number of local neighbors for experiments. In the experiment, the number of neighborhood points directly affects the calculation speed, so the appropriate neighborhood points have an important impact on the method.

Table 5
Sentence Similarity.

Sentences	Similarity	Source
(Sentence1, Sentence2)	1.2	Real score
(Sentence3, Sentence4)	2.8	
(Sentence1, Sentence2)	0.78	SBERT
(Sentence3, Sentence4)	0.64	
(Sentence1, Sentence2)	0.12	Our method
(Sentence3, Sentence4)	0.71	

Table 6
Sentence Similarity.

Sentences	Similarity	Source
(Sentence1, Sentence2)	3.8	Real score
(Sentence3, Sentence4)	4.2	
(Sentence1, Sentence2)	0.95	
(Sentence3, Sentence4)	0.88	SBERT
(Sentence1, Sentence2)	0.67	
(Sentence3, Sentence4)	0.82	

4.6. Case study

In this section, we conduct experiments on the STS13 dataset, using our method to re-embed the sentences contained in the STS13 and then calculate the cosine similarity of sentence pairs. The results of sentence pairs obtained by using the SBERT are also listed, as shown in Table 5 and Table 6.

In Table 5, Sentence1=“an agent expends effort towards achieving a goal. alternatively, a salient entity involved in the goal can be expressed in place of a goal expression.”, Sentence2=“ctivity directed toward making or doing something.”, Sentence3=“cthis frame has to do with scientific taxonomy. the lexical units in this frame include the seven major classifications in which organisms are grouped based on common biological characteristics. these classifications are labeled rank in this frame. The members or subtype of the rank is often expressed with the use of the target-denoting noun.”, Sentence4=“(biology) taxonomic group containing one or more families”. In Table 6, Sentence1=“nato soldier killed in afghan attack”, Sentence2=“nato soldier killed in s. afghanistan”, Sentence3=“castro celebrates 86th birthday monday”, Sentence4=“fidel castro celebrates 86th birthday”. According to the experimental results in Table 5 and Table 6, We can find that the results of real score and SBERT are opposite. The reason is that the existing sentence representation methods do not take into account the geometric structure information between sentences, resulting in the inconsistency between the similarity of sentences in Euclidean metric space and human experience judgment. To address this problem, our method aims to provide an effective sentence representation based on manifold learning, From Table 5 and Table 6, it can be seen that the sentence vector similarity obtained after improving SBERT model by manifold learning is consistent with the real score of sentence pairs.

5. Conclusions

In this paper, we study sentence representation refinement problem by utilizing manifold learning algorithms. We represent sentence BERT in manifold space. Different from the existing sentence embedding methods based on Euclidean space. We use the nearest neighbor graph theory to construct the local geometric structure between sentence vectors, and describe the similarity measurement relationship between sentences. Our sentence re-embedding model can accurately represent the real semantic information between sentences. We conduct several experiments on semantic textual similarity tasks, text classification tasks and document clustering. The results obtained in these three evaluation tasks suggest that our proposed methods consistently perform well for refining sentence representations.

CRedit authorship contribution statement

Yonghe Chu: Data curation, Writing – original draft. **Heling Gao:** Supervision. **Yufeng Diao:** Supervision. **Hongfei Lin:** Supervision.

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence

the work reported in this paper.

Data availability

No data was used for the research described in the article.

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